Ex. 9.1 Q36. $a_n = \frac{n+3}{n^2+5n+6}$

Solution.

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n+3}{(n+2)(n+3)} = \lim_{n\to\infty} \frac{1}{n+2} = 0.$$

The sequence $\{a_n\}$ converges and its limit is 0.

Ex. 9.1 Q48. $a_n = n\pi\cos(n\pi)$

Solution 1. Note that

$$\begin{aligned} a_n + a_{n+1} &= n\pi\cos(n\pi) + (n+1)\pi\cos((n+1)\pi) \\ &= n\pi(-1)^n + (n+1)\pi(-1)^{n+1} \\ &= n\pi(-1)^n - (n+1)\pi(-1)^n \\ &= -\pi(-1)^n \end{aligned}$$

for all $n \in \mathbb{N}$. If $\{a_n\}$ converges to L, then $\{a_{n+1}\}$ also converges to L. So

$$\lim_{n\to\infty}(a_n + a_{n+1}) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} a_{n+1} = L + L = 2L.$$

Hence $\{a_n + a_{n+1}\}$ converges, but $a_n + a_{n+1} = -\pi(-1)^n$, which diverges. Therefore $\{a_n\}$ diverges.

Solution 2. Since $\cos(n\pi) = (-1)^n$ for all $n \in \mathbb{N}$, we have

$$a_n = n\pi(-1)^n.$$

Since $\{a_n\}$ oscillates between large positive and negative numbers, it does not converge to a real number. Hence $\{a_n\}$ diverges.

Ex. 9.1 Q58. $a_n = \left(1 - \frac{1}{n}\right)^n$

Solution.

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = e^{-1} = \frac{1}{e}.$$

The sequence $\{a_n\}$ converges and its limit is $\frac{1}{e}$.

Remark.

$$\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

for all $x \in \mathbb{R}$.

Ex. 9.1 Q70. $a_n = \frac{n!}{2^n \cdot 3^n}$

Solution 1. For $n \geq 6$, we have

$$a_n = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdots n}{6^n} = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}{6^5} \cdot \frac{6}{6} \cdot \frac{7}{6} \cdots \frac{n}{6} \geq \frac{5!}{6^5} \left(\frac{7}{6}\right)^{n-6}.$$

Since $|\frac{7}{6}| > 1$, we have

$$\lim_{n\to\infty} \left(\frac{7}{6}\right)^{n-6} = \infty.$$

Hence $a_n \to \infty$. We conclude that the sequence $\{a_n\}$ diverges.

Solution 2. Since

$$\lim_{n\to\infty} \frac{6^n}{n!} = 0,$$

and $\frac{6^n}{n!} > 0$ for all $n \in \mathbb{N}$, we get

$$\lim_{n\to\infty} \frac{n!}{6^n} = \infty.$$

Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n!}{6^n} = \infty.$$

We conclude that the sequence $\{a_n\}$ diverges.

Ex. 9.1 Q92. $a_n = \frac{1}{\sqrt{n}} \arctan n$

Solution. Since

$$0 \leq \arctan n \leq \frac{\pi}{2}$$

for all $n \in \mathbb{N}$, we have

$$0 \leq a_n = \frac{1}{\sqrt{n}} \arctan n \leq \frac{\pi}{2\sqrt{n}}.$$

Since $\frac{\pi}{2\sqrt{n}} \to 0$, by Sandwich Theorem, we get

$$\lim_{n\to\infty} a_n = 0.$$

The sequence $\{a_n\}$ converges and its limit is 0.

Ex. 9.1 Q94. $a_n = \sqrt[n]{n^2+n}$

Solution. Write

$$a_n = \sqrt[n]{n^2+n} = \sqrt[n]{n^2\left(1+\frac{1}{n}\right)} = n^{2/n}\left(1+\frac{1}{n}\right)^{1/n}.$$

Since $\sqrt[n]{n} \to 1$, we have $n^{2/n} \to 1$. Also,

$$1 \leq \left(1+\frac{1}{n}\right)^{1/n} \leq 2^{1/n}.$$

Since $2^{1/n} \to 1$, by Sandwich Theorem, we get

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{1/n}.$$

Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} n^{2/n} \cdot \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{1/n} = 1 \cdot 1 = 1.$$

The sequence $\{a_n\}$ converges and its limit is 1.

Ex. 9.2 Q11. $\sum_{n=0}^{\infty}\left(\frac{5}{2^n}+\frac{1}{3^n}\right)$

Solution. Note that

$$\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n = \frac{1}{1-\frac{1}{2}} = 2,$$

$$\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n = \frac{1}{1-\frac{1}{3}} = \frac{3}{2}.$$

So

$$\begin{aligned}\sum_{n=0}^{\infty}\left(\frac{5}{2^n}+\frac{1}{3^n}\right) &= \sum_{n=0}^{\infty}\frac{5}{2^n}+\sum_{n=0}^{\infty}\frac{1}{3^n}\\ &= 5\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n+\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n\\ &= 5(2)+\frac{3}{2}\\ &= \frac{23}{2}.\end{aligned}$$

The series converges and its sum is $\frac{23}{2}$.

Ex. 9.2 Q35. $\sum_{n=1}^{\infty} \cos \frac{1}{n}$

Solution. Let $a_n = \cos \frac{1}{n}$. Since $\frac{1}{n} \to 0$ and $\cos x$ is continuous at 0, we have

$$\lim_{n\to\infty} a_n = \cos\left(\lim_{n\to\infty} \frac{1}{n}\right) = \cos 0 = 1.$$

Since $\lim_{n\to\infty} a_n \neq 0$, by the n-th Term Test, the series $\sum_{n=1}^{\infty} \cos \frac{1}{n}$ diverges.

Ex. 9.2 Q48. $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2}$

Solution. Let $s_n = \sum_{k=1}^{n} \frac{2k+1}{k^2(k+1)^2}$ for $n \in \mathbb{N}$. Then

$$\begin{aligned}
s_n &= \sum_{k=1}^{n} \frac{(k+1)^2 - k^2}{k^2(k+1)^2} \\
&= \sum_{k=1}^{n} \left(\frac{1}{k^2} - \frac{1}{(k+1)^2} \right) \\
&= \left(1 - \frac{1}{2^2}\right) + \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \cdots + \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) \\
&= 1 - \frac{1}{(n+1)^2}.
\end{aligned}$$

Hence

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \left(1 - \frac{1}{(n+1)^2}\right) = 1.$$

The series converges and its sum is 1.

Ex. 9.2 Q58. $\sum_{n=0}^{\infty} \frac{\cos n\pi}{5^n}$

Solution. Since $\cos n\pi = (-1)^n$ for all $n \in \mathbb{N}$, we have

$$\sum_{n=0}^{\infty} \frac{\cos n\pi}{5^n} = \sum_{n=0}^{\infty} \left(-\frac{1}{5}\right)^n.$$

Since $\left|-\frac{1}{5}\right| < 1$, the series converges and its sum is

$$\sum_{n=0}^{\infty} \frac{\cos n\pi}{5^n} = \frac{1}{1-\left(-\frac{1}{5}\right)} = \frac{1}{1+\frac{1}{5}} = \frac{5}{6}.$$

The series converges and its sum is $\frac{5}{6}$.

Ex. 9.2 Q68. $\sum_{n=1}^{\infty} \frac{2^n+4^n}{3^n+4^n}$

Solution. Let

$$a_n = \frac{2^n + 4^n}{3^n + 4^n}.$$

Then

$$a_n = \frac{\left(\frac{1}{2}\right)^n + 1}{\left(\frac{3}{4}\right)^n + 1}.$$

Since $0 < \frac{1}{2} < 1$ and $0 < \frac{3}{4} < 1$, we have $\left(\frac{1}{2}\right)^n \to 0$ and $\left(\frac{3}{4}\right)^n \to 0$. Thus

$$\lim_{n\to\infty} a_n = \frac{0+1}{0+1} = 1.$$

Since $\lim_{n\to\infty} a_n \neq 0$, by the n-th Term Test, the series $\sum_{n=1}^{\infty} \frac{2^n+4^n}{3^n+4^n}$ diverges.

Ex. 9.2 Q70. $\sum_{n=1}^{\infty} \ln\left(\frac{n}{2n+1}\right)$

Solution. Let

$$a_n = \ln\left(\frac{n}{2n+1}\right).$$

Note that

$$\lim_{n\to\infty} \frac{n}{2n+1} = \lim_{n\to\infty} \frac{1}{2+\frac{1}{n}} = \frac{1}{2}.$$

Since $\ln x$ is continuous at $\frac{1}{2}$, we get

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \ln\left(\frac{n}{2n+1}\right) = \ln\left(\lim_{n\to\infty} \frac{n}{2n+1}\right) = \ln\frac{1}{2} \neq 0.$$

By the n-th Term Test, the series $\sum_{n=1}^{\infty} \ln\left(\frac{n}{2n+1}\right)$ diverges.

Ex. 9.3 Q30. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$

Solution. Let $f : [1, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{1}{\sqrt{x}(\sqrt{x}+1)} \quad \text{for all } x \in [1, \infty).$$

Then

$$\begin{aligned}
\int_1^{\infty} \frac{1}{\sqrt{x}(\sqrt{x}+1)}\, dx &= \lim_{b\to\infty} \int_1^{b} \frac{1}{\sqrt{x}(\sqrt{x}+1)}\, dx \\
&= 2 \lim_{b\to\infty} \int_1^{\sqrt{b}} \frac{1}{u+1}\, du \qquad [u = \sqrt{x}] \\
&= 2 \lim_{b\to\infty} [\ln(u+1)]_1^{\sqrt{b}} \\
&= 2 \lim_{b\to\infty} \left(\ln(\sqrt{b}+1) - \ln 2\right) \\
&= \infty.
\end{aligned}$$

So $\int_1^{\infty} \frac{1}{\sqrt{x}(\sqrt{x}+1)}\, dx$ diverges. Since f is continuous, positive and decreasing on $[1, \infty)$, by the Integral Test, the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$ diverges.

Ex. 9.3 Q41. $\sum_{n=2}^{\infty} \frac{\sqrt{n+2}-\sqrt{n+1}}{\sqrt{n+1}\sqrt{n+2}}$

Solution. Let $s_n = \sum_{k=2}^{n} \frac{\sqrt{k+2}-\sqrt{k+1}}{\sqrt{k+1}\sqrt{k+2}}$ for $n \in \mathbb{N}$. Then

$$\begin{aligned} s_n &= \sum_{k=2}^{n} \left(\frac{1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+2}} \right) \\ &= \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} \right) + \left(\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{5}} \right) + \cdots + \left(\frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n+2}} \right) \\ &= \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{n+2}}. \end{aligned}$$

Hence

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{n+2}} \right) = \frac{1}{\sqrt{3}}.$$

The series converges and its sum is $\frac{1}{\sqrt{3}}$.

Ex. 9.3 Q43. $\sum_{n=1}^{\infty} \frac{8\tan^{-1} n}{1+n^2}$

Solution. Let $f : [1, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{8\tan^{-1} x}{1+x^2} \quad \text{for all } x \in [1, \infty).$$

Then

$$\begin{aligned}
\int_1^{\infty} \frac{8\tan^{-1} x}{1+x^2}\, dx &= \lim_{b\to\infty} \int_1^{b} \frac{8\tan^{-1} x}{1+x^2}\, dx \\
&= \lim_{b\to\infty} \int_{\pi/4}^{\tan^{-1} b} 8u\, du \qquad [u = \tan^{-1} x] \\
&= 4 \lim_{b\to\infty} \left[u^2\right]_{\pi/4}^{\tan^{-1} b} \\
&= 4 \lim_{b\to\infty} \left((\tan^{-1} b)^2 - \left(\frac{\pi}{4}\right)^2\right) = 4\left(\left(\frac{\pi}{2}\right)^2 - \left(\frac{\pi}{4}\right)^2\right) = \frac{3\pi^2}{4}.
\end{aligned}$$

So $\int_1^{\infty} \frac{8\tan^{-1} x}{1+x^2}\, dx$ converges. Since f is continuous, positive and decreasing on $[1, \infty)$, by the Integral Test, the series $\sum_{n=1}^{\infty} \frac{8\tan^{-1} n}{1+n^2}$ converges.